

EX:02	Design and Implement LLM-Based Applications for Question Answering, Text Summarization, and Sentiment Analysis
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Aim:

To design and implement Large Language Model (LLM)-based applications for **Question Answering, Text Summarization, and Sentiment Analysis** using the **Google Gemini API**, and evaluate their performance on real-world textual data.

Procedure:

1. Install the required Python libraries (google-genai).
2. Obtain a Google Gemini API key.
3. Initialize the Gemini client using the API key.
4. Select real-world textual data (news article, product review, or paragraph).
5. Implement a **Question Answering** application by asking questions about the text.
6. Implement a **Text Summarization** application to generate a concise summary.
7. Implement a **Sentiment Analysis** application to classify the text as Positive, Negative, or Neutral with an explanation.
8. Execute the applications using the Gemini model.
9. Compare the generated outputs with the expected results.
10. Record the observations and evaluate the model's performance.

Code:

```
from google import genai

# Initialize Gemini client
client = genai.Client(api_key="YOUR_GEMINI_API_KEY")

text = """
I bought this laptop two months ago hoping it would last through my long
working days. The display is bright and the keyboard feels comfortable,
but the battery barely survives three hours. The fan becomes very loud
during video calls, and the charger stopped working after six weeks.
Customer support took nearly two weeks to respond, which was disappointing.
"""

# -----
# 1. Question Answering
# -----
question = "What problems did the customer face with the laptop?"

qa_prompt = f"""
Read the following text and answer the question.

Text:
```

```
{text}
```

Question:

```
{question}
```

```
"""
```

```
qa_response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents=qa_prompt
)
```

```
print("="*60)
print("QUESTION ANSWERING")
print("="*60)
print(qa_response.text)
```

```
# -----
```

```
# 2. Text Summarization
```

```
# -----
```

```
summary_prompt = f"""
```

```
Summarize the following text in 3-4 sentences.
```

```
{text}
```

```
"""
```

```
summary_response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents=summary_prompt
)
```

```
print("\n"+"="*60)
print("TEXT SUMMARIZATION")
print("="*60)
print(summary_response.text)
```

```
# -----
```

```
# 3. Sentiment Analysis
```

```
# -----
```

```
sentiment_prompt = f"""
```

```
Analyze the sentiment of the following text.
```

Classify it as:

- Positive
- Negative
- Neutral

Also provide a brief explanation.

Text:

```
{text}
```

```
"""
```

```

sentiment_response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents=sentiment_prompt
)

print("\n"+"="*60)
print("SENTIMENT ANALYSIS")
print("="*60)
print(sentiment_response.text)

```

Output:

```

=====
QUESTION ANSWERING
=====

```

The customer faced the following problems with the laptop:

- * The battery lasts only about three hours
- * The fan becomes very loud during video calls
- * The charger stopped working after six weeks
- * Customer support took nearly two weeks to respond

```

=====
TEXT SUMMARIZATION
=====

```

The customer purchased a laptop two months ago expecting it to support long working days. Although the display is bright and the keyboard is comfortable, the battery lasts only about three hours and the fan is very noisy during video calls. The charger also failed after six weeks, and customer support took almost two weeks to reply, which left the customer disappointed.

```

=====
SENTIMENT ANALYSIS
=====

```

****Sentiment:** Negative**

****Explanation:****

The text begins with a hopeful expectation, but most of it describes problems with the product and the service. Although the review briefly praises the "bright" display and "comfortable" keyboard, these positives are outweighed by complaints such as the battery that "barely survives three hours," a "very loud" fan, a charger that "stopped working after six weeks," and slow customer support. The review ends with the word "disappointing," which clearly shows dissatisfaction. The overall tone is therefore negative, with only minor positive remarks.

Result:

The LLM was successfully used to implement Question Answering, Text Summarization, and Sentiment Analysis applications. The model correctly listed all four problems in the review, produced a faithful summary, and classified the review as Negative with a sound explanation. The experiment demonstrates that LLMs are effective for a variety of natural language processing tasks and can produce high-quality results with simple prompts.